FIELD AND SERVICE ROBOTICS

Robot Kinematics, Actuation and Nonholonomic Constraints

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GitHub repository

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Chapter 1

Full Actuation of ATLAS: Standing vs Aerial Phase

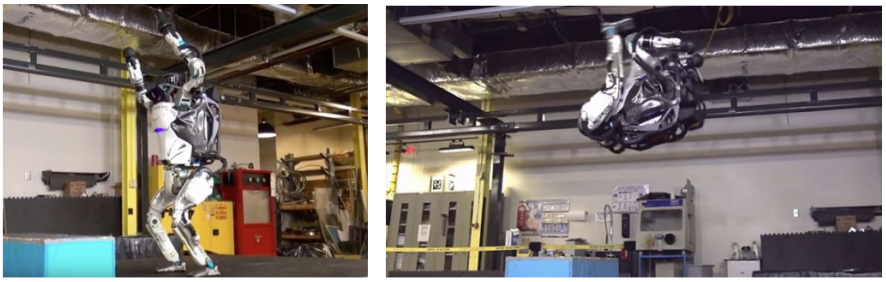


Figure 1.1: ATLAS configurations

a) **While standing, ATLAS is fully actuated.**
True. When ATLAS is standing and maintains stable contact with the ground, suppose that there is sufficient friction which prevents foot slippage, the contact provides a reaction point through which the robot can generate external forces and torques. Under these conditions, there always exists a suitable combination of inputs τ at the robot's joints that allows independent control over all generalized coordinates. In particular, by appropriately choosing τ, the robot can impose arbitrary accelerations on each generalized coordinate, including the spatial position of its center of mass. Thus, the system is **fully actuated**.

b) **While doing backflip, ATLAS is fully actuated.**
False. When ATLAS leaves the ground during a backflip, the situation changes. In this aerial phase, it still controls its internal joints but lacks actuators to directly control the ballistic trajectory of its center of mass. Although it can redistribute angular momentum internally by moving its limbs, it cannot generate external forces or torques to arbitrarily adjust its global motion. Therefore, during the aerial phase of a backflip, there are

desired accelerations that cannot be achieved. This makes ATLAS **underactuated** in flight, unable to exert full control over its spatial trajectory without external support.

Chapter 2

Degrees of Freedom and Configuration Space topology

Analysis of Degrees of Freedom and Configuration Space Topologies

I analyze the degrees of freedom (DoFs) and configuration space topologies of two spatial mechanisms using Grübler's formula:

$$\text{DoFs} = m(N - 1 - J) + \sum_{i=1}^{J} f_i \tag{2.1}$$

where m is the number of degrees of freedom of a rigid body in space ($m = 6$ for spatial mechanisms), N is the number of links, J is the number of joints, and f_i is the degree of freedom provided by each joint.

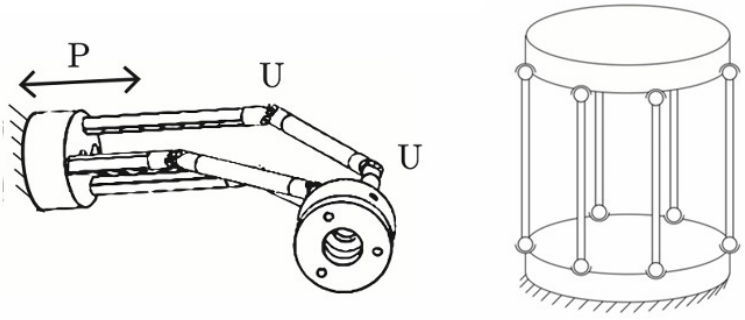


Figure 2.1: Spatial mechanisms

First Mechanism: Surgical Manipulator

The surgical manipulator (on the left), developed at the National University of Singapore, consists of three identical parts connecting a fixed base to a moving

platform. Each part includes two universal joints (each providing 2 DoFs) and one prismatic joint (providing 1 DoF). Thus, the total number of joints is $J = 3 \times 3 = 9$ and the total number of links is $N = 1$ (ground) + 1 (platform) + 6 (segments) = 8. Applying Grübler's formula yields:

$$\begin{aligned} \text{DoFs} &= 6(8 - 1 - 9) + 15 && (2.2)\\ &= 6(-2) + 15 && (2.3)\\ &= 3 && (2.4) \end{aligned}$$

The manipulator thus possesses exactly 3 degrees of freedom. Its configuration space topology is I x I x I.

Second Mechanism: 6-Bar Spatial Structure

The second mechanism (on the right) comprises a fixed base, a moving platform, and six identical bars, each connected by spherical joints (each providing 3 DoFs) at both ends. Therefore, the total number of joints is $J = 6 \times 2 = 12$, and the total number of links remains $N = 1$ (ground) + 1 (platform) + 6 (bars) = 8. Using Grübler's formula:

$$\begin{aligned} \text{DoFs} &= 6(8 - 1 - 12) + 12 \times 3 && (2.5)\\ &= 6(-5) + 36 && (2.6)\\ &= 6 && (2.7) \end{aligned}$$

This mechanism thus has 6 degrees of freedom. Given the presence of dependent constraints in the mechanism, Grübler's formula provides only a lower bound for the system's degrees of freedom.
It is therefore concluded that the system has **9** degrees of freedom. In particular, the upper platform is capable of translating horizontally while simultaneously rotating and translating about the vertical axis. Each bar can rotate around its own axis. Consequently, its configuration space topology is $\mathbb{T}^6 \times I^2 \times S^1$.

Chapter 3

Assessment of actuation properties

a. A car with inputs the steering angle and the throttle is underactuated.

True. The car cannot produce arbitrary accelerations in all planar directions using only steering and throttle inputs. Specifically, it lacks the ability to directly induce motion in the direction orthogonal to its forward-facing axis, which prevents lateral displacement. Consequently, this control limitation classifies the vehicle as underactuated. Plus, the system has three DoFs given by the planar rigid body chassis but only two control inputs, this mismatch ($\dim(\tau) < \dim(q)$) is a defining feature of underactuated systems.

b. The KUKA youBot system on the slides is fully actuated.

True. The KUKA youBot is composed of an omnidirectional mobile base equipped with four mecanum wheels, each driven independently. In addition, there is a robotic arm mounted on top, with a motor for each joint. The wheel configuration permits the base to translate and rotate freely on the plane. This full controllability over planar motion, together with the joint level control of the manipulator, renders the overall system fully actuated.

c. The hexarotor system with co-planar propellers is fully actuated.

False. Despite having six rotors, their co-planar arrangement restricts the thrust vectors to a single direction, perpendicular to the rotor plane. All forces generated are thus aligned along the vertical axis. To achieve horizontal motion, the drone must tilts itself to project some of its thrust force laterally. Without tilting, lateral accelerations are impossible. The addition of more co-planar rotors wouldn't change this limitation. In conclu-

sion, a hexarotor with co-planar propellers is an **underactuated system** in which the direction of the total force that can be applied is constrained along the vertical axis of the drone and it is not possible to generate arbitrary accelerations in all directions in space

d. The KUKA iiwa 7-DOF robot is redundant and it cannot be underactuated because we know that all redundant systems are not.

False. The KUKA iiwa has more degrees of freedom than minimally required for certain tasks, making it redundant. This doesn't guarantee that it is fully actuated in the dynamic sense. Redundancy pertains to the kinematic level. Underactuation, however, is a dynamic concept that deals with whether the available control inputs can generate arbitrary accelerations across all generalized coordinates. These two properties are not mutually exclusive, nor do they imply one another. It is entirely possible for a system to be kinematically redundant yet dynamically underactuated, depending on its actuation structure and task.

Chapter 4

Distributions

(a) Distribution Δ_1

$$\Delta_1 = \left\{ \begin{bmatrix} -3x_2 \\ 1 \\ -1 \end{bmatrix} \right\}, \quad U \in \mathbb{R}^3$$

Involutivity Check

Involutive? Yes, since any 1-dimensional distribution is involutive, [f,f]=0, and the zero vector belongs to any distribution by default.

Annihilator of Δ_1

The **annihilator** of the distribution Δ_1 is defined as the set of covectors that vanish on all vectors of the distribution. Its dimension is the difference between the dimension of the vector space and the dimension of the distribution:

$$\dim(\Delta_1^{\perp}(x)) = n - \dim(\Delta_1(x)) = 3 - 1 = 2.$$

Explicitly, the annihilator can be expressed as:

$$\Delta_1^{\perp}(x) = \left\{ \omega^* \in \mathbb{R}^{3*} : \langle \omega^*, \nu \rangle = 0, \ \forall \nu \in \Delta_1(x) \right\},$$

where

$$\omega^* = \begin{bmatrix} \omega_1 & \omega_2 & \omega_3 \end{bmatrix}.$$

To find $\Delta_1^{\perp}$, impose the condition:

$$\omega^* F = \mathbf{0}_3,$$

where F is the matrix representing the vector fields of Δ_1, and $\mathbf{0}_3$ is the zero vector in $\mathbb{R}^3$. This yields:

$$\begin{bmatrix}\omega_1 & \omega_2 & \omega_3\end{bmatrix}\begin{bmatrix}-3x_2\\1\\-1\end{bmatrix}=0 \quad \Rightarrow \quad -3x_2\omega_1+\omega_2-\omega_3=0.$$

Thus, ω_2 can be expressed as:

$$\omega_2=\omega_3+3x_2\omega_1.$$

Choosing independent values for ω_1 and ω_3, we obtain:

- For $\omega_1=0$, $\omega_3=1$:
$$\omega_1^*=\begin{bmatrix}0 & 1 & 1\end{bmatrix}.$$
- For $\omega_1=1$, $\omega_3=0$:
$$\omega_2^*=\begin{bmatrix}1 & 3x_2 & 0\end{bmatrix}.$$

Therefore, the annihilator of Δ_1 is:

$$\Delta_1^{\perp}(x)=\operatorname{span}\left\{\begin{bmatrix}0 & 1 & 1\end{bmatrix},\ \begin{bmatrix}1 & 3x_2 & 0\end{bmatrix}\right\}.$$

(b) Distribution Δ_2

Consider the distribution Δ_2 defined by:

$$\Delta_2=\operatorname{span}\left\{\begin{bmatrix}-1\\0\\x_3\end{bmatrix},\quad \begin{bmatrix}x_2\\-3\\x_1\end{bmatrix}\right\},\quad U\in\mathbb{R}^3,\quad \alpha=3.$$

Non-singularity Verification

The distribution is non-singular if the matrix

$$F=\begin{bmatrix}-1 & x_2\\0 & -3\\x_3 & x_1\end{bmatrix}$$

has rank equal to 2. Clearly, this is satisfied, implying:

$$\dim(\Delta_2)=2=d\quad \forall x\in U,\quad U\in\mathbb{R}^3.$$

Thus, the distribution is non-singular.

Involutivity Check

To verify involutivity, we compute the Lie bracket $[f_1(x), f_2(x)]$:

$$[f_1(x), f_2(x)] = \frac{\partial f_2}{\partial x} f_1(x) - \frac{\partial f_1}{\partial x} f_2(x).$$

Explicit computation yields:

$$[f_1(x), f_2(x)] = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} -1 \\ 0 \\ x_3 \end{bmatrix} - \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_2 \\ -3 \\ x_1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ -1 - x_1 \end{bmatrix}.$$

Check linear dependence by adding this vector as a new column to matrix F:

$$\tilde{F} = \begin{bmatrix} -1 & x_2 & 0 \\ 0 & -3 & 0 \\ x_3 & x_1 & -1 - x_1 \end{bmatrix}, \quad \det(\tilde{F}) = 3(-1 - x_1).$$

Thus, the determinant is zero only for $x_1 = -1$, indicating involutivity only on this plane.

- **Therefore, the distribution Δ_2 is generally not involutive in U, except on the plane $x_1 = -1$.**

Annihilator of Δ_2

The annihilator exists with dimension:

$$\dim(\Delta_2^{\perp}(x)) = 3 - \dim(\Delta_2(x)) = 3 - 2 = 1.$$

Define:

$$\Delta_2^{\perp}(x) = \left\{ \omega^* \in \mathbb{R}^{3*} : \langle \omega^*, \nu \rangle = 0, \forall \nu \in \Delta_2(x) \right\}, \quad \omega^* = [\omega_1 \ \omega_2 \ \omega_3].$$

Solve:

$$\omega^* F = 0 \Rightarrow \begin{cases} -\omega_1 + \omega_3 x_3 = 0 \\ \omega_1 x_2 - 3\omega_2 + \omega_3 x_1 = 0 \end{cases} \Rightarrow \begin{cases} \omega_1 = \omega_3 x_3 \\ \omega_2 = \frac{\omega_3 (x_2 x_3 + x_1)}{3} \end{cases}$$

Choosing $\omega_3 = 3$:

$$\omega^* = [3x_3 \quad x_1 + x_2 x_3 \quad 3].$$

- **Thus, the annihilator is:**

$$\Delta_2^{\perp}(x) = \text{span}\{[3x_3 \quad x_1 + x_2 x_3 \quad 3]\}.$$

(c) Distribution Δ_3

Consider the distribution Δ_3 defined by:

$$\Delta_3 = \text{span}\left\{ \begin{bmatrix} 2x_3 \\ 1 \\ 0 \end{bmatrix}, \quad \begin{bmatrix} -2x_2 \\ x_1 \\ -1 \end{bmatrix} \right\}, \quad U \in \mathbb{R}^3.$$

Non-singularity Verification

First, we verify the distribution is non-singular by checking the rank of the matrix:

$$F = \begin{bmatrix} 2x_3 & -2x_2 \\ 1 & x_1 \\ 0 & -1 \end{bmatrix}.$$

We observe clearly that:

$$\text{rank}(F) = 2,$$

which implies:

$$\dim(\Delta_3) = 2 = d \quad \forall x \in U, \quad U \in \mathbb{R}^3.$$

Thus, the distribution is non-singular.

Involutivity Check

We test involutivity by calculating the Lie bracket $[f_1(x), f_2(x)]$:

$$[f_1(x), f_2(x)] = \frac{\partial f_2}{\partial x} f_1(x) - \frac{\partial f_1}{\partial x} f_2(x).$$

The explicit computation yields:

$$[f_1(x), f_2(x)] = \begin{bmatrix} 0 & -2 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 2x_3 \\ 1 \\ 0 \end{bmatrix} - \begin{bmatrix} 0 & 0 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} -2x_2 \\ x_1 \\ -1 \end{bmatrix} = \begin{bmatrix} 0 \\ 2x_3 \\ 0 \end{bmatrix}.$$

Check linear dependence by adding this result as the third column of the matrix F:

$$\tilde{F} = \begin{bmatrix} 2x_3 & -2x_2 & 0 \\ 1 & x_1 & 2x_3 \\ 0 & -1 & 0 \end{bmatrix}, \quad \det(\tilde{F}) = 4x_3^2.$$

The determinant vanishes only for $x_3 = 0$, indicating that involutivity is restricted to this condition.

- **Therefore, Δ_3 is generally not involutive in U, since the condition**

  $$[f_1(x), f_2(x)] \in \Delta_3(x), \quad \forall x \in U \subseteq \mathbb{R}^3$$

  is not satisfied everywhere.

Annihilator of Δ_3

The **annihilator** of the distribution Δ_3 exists, and its dimension is determined as follows:

$$\dim(\Delta_3^{\perp}(x)) = n - \dim(\Delta_3(x)) = 3 - 2 = 1.$$

Formally, the annihilator is defined by:

$$\Delta_3^{\perp}(x) = \left\{\omega^* \in \mathbb{R}^{3*} : \langle \omega^*, \nu \rangle = 0,\ \forall \nu \in \Delta_3(x)\right\},$$

where the generic covector is expressed as:

$$\omega^* = \begin{bmatrix}\omega_1 & \omega_2 & \omega_3\end{bmatrix}.$$

To explicitly compute the annihilator $\Delta_3^{\perp}$, we impose the orthogonality condition:

$$\omega^* F = 0,$$

where the matrix F contains the generating vector fields of the distribution Δ_3. Thus, we have:

$$\begin{bmatrix}\omega_1 & \omega_2 & \omega_3\end{bmatrix} \begin{bmatrix} 2x_3 & -2x_2 \\ 1 & x_1 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix}0 & 0\end{bmatrix},$$

which yields the following system of equations:

$$\begin{cases} 2x_3\,\omega_1 + \omega_2 = 0 \\ -2x_2\,\omega_1 + x_1\,\omega_2 - \omega_3 = 0 \end{cases}$$

Solving the above system, we obtain:

$$\begin{cases} \omega_2 = -2x_3\,\omega_1 \\ \omega_3 = -2\omega_1(x_2 + x_1x_3) \end{cases}$$

Thus, the general covector form is:

$$\omega^* = \begin{bmatrix}\omega_1 & -2x_3\,\omega_1 & -2\omega_1(x_2 + x_1x_3)\end{bmatrix}.$$

Choosing, for instance, $\omega_1 = \frac{1}{2}$ to avoid trivial solutions, we obtain an explicit representative covector field:

$$\omega^* = \begin{bmatrix}\frac{1}{2} & -x_3 & -x_2 - x_1x_3\end{bmatrix}.$$

- **Therefore, the annihilator of the distribution Δ_3 is given by:**

$$\Delta_3^{\perp}(x) = \operatorname{span}\left\{\begin{bmatrix}\frac{1}{2} & -x_3 & -x_2 - x_1x_3\end{bmatrix}\right\}.$$

Chapter 5

Accessibility rank condition and Pfaffian constraints

Consider a system subject to Pfaffian constraints:

$$A\dot{q} = 0,$$

where $A \in \mathbb{R}^{k\times n}$ is a constant matrix, independent of the configuration $q \in \mathbb{R}^n$. We aim to show that, in this form, the constraints are integrable and thus holonomic. Each row of A represents an independent constraint. Since A is constant, the constraints are fixed and define a subspace of admissible velocities. The set of such velocities defines the *kinematic distribution*:

$$\Delta = \operatorname{span}\left\{g_j(q) \in \mathbb{R}^n \mid Ag_j(q) = 0,\ j = 1, \ldots, m\right\}.$$

A generic vector belonging to the distribution can be expressed as:

$$\dot{q} = \sum_{j=1}^{m} g_j(q)u_j = G(q)u,$$

where $u \in \mathbb{R}^m$ is the input. This is the *driftless affine kinematic model*. Now consider the *accessibility distribution* Δ_A, generated by the vector fields g_j and their Lie brackets. Since A is constant, the g_j do not depend on q, and all Lie brackets vanish:

$$[g_i, g_j] = 0 \quad \forall i, j \Rightarrow \Delta_A = \Delta.$$

The dimension of the distribution is:

$$\dim(\Delta_A) = \dim(\Delta) = m = n - \operatorname{rank}(A).$$

Assuming the k constraints are independent, $\operatorname{rank}(A) = k$, and thus:

$$m = n - k < n.$$

Since $\dim(\Delta_A) < n$, the system is *not accessible* and *not controllable*. However, the involutivity and constancy of the distribution imply that the Pfaffian constraints are *completely integrable*, hence holonomic.

Chapter 6

Raibert's hooper robot

Consider the Raibert's hopper robot, subject to the Pfaffian constraint:

$$(I + m(d+l)^2)\,\dot{\theta} + m(d+l)^2\,\dot{\psi} = 0,$$

with generalized coordinates:

$$q = \begin{bmatrix} \theta \\ \psi \\ I \end{bmatrix}.$$

The kinematic model was derived and the constraint was verified to be non-integrable, meaning the system is completely non-holonomic. The Pfaffian matrix A was used in MATLAB to compute the null space, obtaining the matrix G whose columns span the kinematic distribution:

$$G = \begin{bmatrix} -\dfrac{m(d+l)^2}{md^2 + 2mdl + ml^2 + I} & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}$$

The first vector field $g_1(q)$ depends on the generalized coordinate I, so the distribution is configuration-dependent. The accessibility distribution was computed from g_1, g_2, and their Lie bracket $[g_1, g_2]$, which turned out to be non-zero:

$$F = \begin{bmatrix} -\dfrac{m(d+l)^2}{I + m(d+l)^2} & 0 & \dfrac{m(2d+2l)}{I + m(d+l)^2} - \dfrac{m^2(d+l)^2(2d+2l)}{(I + m(d+l)^2)^2} \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

The rank of F is:

$$\operatorname{rank}(F) = 3$$

Since $\dim(\Delta_A) = 3 = n$, the accessibility rank condition is satisfied. The system is then controllable. Therefore, the system is completely non-holonomic. **The MATLAB's code, where is possible to display the matrix F and its rank, is attached in Moodle.**